Eigen-Séta: Public-Key Encryption from a Secret Endomorphism of the Public Curve

Design note and toy-prototype report

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Status. This document describes a proposed scheme, a SageMath prototype that runs its complete KeyGen / Encaps / Decaps path at toy parameters, and the security analysis available today. Security is **not established**: the scheme rests on two assumptions that have no reduction to a studied problem, the prototype exercises only special-key variants at 32–66-bit primes, and all parameter figures are derived estimates, not measurements. Nothing here should be used to protect data.

Abstract

We propose a public-key encryption scheme whose decryption key is a secret endomorphism θ of the public supersingular curve E_A, i.e. a piece of $\mathrm{End}(E_A)$. The public key is E_A together with an eigenbasis (P, Q) of θ on the 2^e-torsion; the ciphertext is the codomain E_C of a secret 3^f-isogeny φ from E_A plus the torsion images $\varphi(P)$, $\varphi(Q)$ masked by a sender-random diagonal matrix. Decryption uses the fact, from Castryck–Vercauteren's generalised lollipop attack, that a mask cancels in the endomorphism $\varphi\theta\hat{\varphi}$ of E_C exactly when it commutes with θ on the torsion: in an eigenbasis every diagonal mask does. The receiver therefore learns $\varphi\theta\hat{\varphi}$ on $E_C[2^e]$ for free, turns it into an explicit isogeny with Kani's lemma, and reads off $\ker\hat{\varphi}$. Compared with SILBE (SAC 2024), the only published encryption scheme built on lollipop decoding, the move from scalar to diagonal masks removes the requirement that the torsion order have at least 2λ prime factors, and the derived prime size drops from 13 013 bits to about 772 bits at $\lambda = 128$ (assuming the two-halves Kani evaluation). We report a prototype, the corrections it forced on the design, a first cryptanalytic experiment, and the open problems.

1. Problem and prior art

Endomorphism-ring hardness. For a supersingular curve $E/\mathbb{F}_{p^2}$, computing $\mathrm{End}(E)$ is equivalent, under probabilistic polynomial-time reductions and now unconditionally, to the isogeny-path, maximal-order, one-endomorphism and arbitrary-degree isogeny problems (Wesolowski 2021; Page–Wesolowski 2024; Herlédan Le Merdy–Wesolowski 2025). Worst-case hardness implies average-case hardness. Best attacks: classical $\tilde{O}(p^{1/2})$ with low memory, heuristic $p^{1/3+o(1)}$ with exponential memory (Wesolowski 2026), quantum $\tilde{O}(p^{1/4})$. Any structure revealed about $\mathrm{End}(E)$ collapses this: one endomorphism gives $|\mathrm{disc}|^{1/4}$ classical and subexponential quantum; a full-rank suborder or an evaluable isogeny from a known-End curve gives quantum polynomial time (Chen et al. 2023; Chen–Petit 2025).

The structural obstruction. Given $\mathrm{End}(E_A)$ alone, computing an isogeny to a fresh curve E_C is equivalent to computing $\mathrm{End}(E_C)$. A receiver holding $\mathrm{End}(E_A)$ therefore cannot decrypt “here is a random curve isogenous to yours”. A ciphertext must carry a hint, and the design question is which hint is useless to everyone, decodable with $\mathrm{End}(E_A)$, and cheap. The seven-open-problems paper of Castryck, De Feo, Galbraith, Kutas, Reijnders and Wesolowski (ePrint 2026/1431) lists encryption from a small variant of EndRing as open Problem 7.

Two known mechanisms. Commutative group actions (CSIDH, SCALLOP, PEGASIS, CORAL) need a public orientation so the sender can act, which exposes a subexponential quantum attack. Torsion hints (the SIDH family: FESTA, QFESTA, POKÉ, PIKE, LIT-SiGamal, Séta, SILBE) reveal torsion images hidden by a subgroup Γ of $\mathrm{GL}_2(\mathbb{Z}/N)$. Unmasked or unipotent hints are broken for everyone once $N^2 > \deg\varphi$ (Castryck–Decru, Maino et al., Robert 2022). Scalar masks (M-SIDH) and diagonal masks (FESTA's CIST problem) only have exponential attacks on generic curves (De Feo–Fouotsa–Panny 2024).

Scheme	Trapdoor	Hint in ciphertext	Why it is where it is
Séta (AC 2021)	curve with a secret endomorphism; Petit's attack as decryptor	exact torsion images	broken 2022: Robert's attack needs no endomorphism
M-SIDH (EC 2023)	key exchange	scalar-masked images, N with ≥ 2λ prime factors	pairing leaks λ^2; needs E_0 of unknown End; 5911-bit p
Castryck–Vercauteren (AC 2023)	attack	lollipop $\varphi'\omega\hat{\varphi}$; mask cancels iff ω commutes with Γ	diagonal masks: only from an eigenbasis; treated as a backdoor
SILBE (SAC 2024)	$\mathrm{End}(E_A)$ via a path from $\mathbb{F}_p$-rational E_0	scalar-masked images (mask = message)	OW-PCA ⇔ M-SIDH problem; p = 13 013 bits; impractical
FESTA / QFESTA / POKÉ	secret isogeny from known-End E_0 + masks	diagonal or scalar masks	key recovery is a torsion-hint problem, not EndRing
Eigen-Séta (this note)	secret $\theta \in \mathrm{End}(E_A)$ with public eigenbasis	sender-random diagonal masks in that basis	lollipop decodable by the owner; CV's backdoor used legitimately

2. The decoding lemma

Let $\varphi: E_A \to E_C$ have degree d, let (P, Q) be a basis of $E_A[N]$, and let the ciphertext reveal $\Lambda = \varphi \circ D$ on $E_A[N]$, where $D \in \Gamma \leq \mathrm{GL}_2(\mathbb{Z}/N)$ is the sender's mask written in the basis (P, Q). For $\theta \in \mathrm{End}(E_A)$ with matrix M on $E_A[N]$, the receiver can compute $\Lambda \circ M \circ (d \cdot \Lambda^{-1}) = \varphi\, D\, M\, D^{-1}\, \hat{\varphi}$ on $E_C[N]$. This equals the lollipop $\psi_\theta := \varphi\theta\hat{\varphi}$ exactly iff DM = MD (Castryck–Vercauteren Lemma 3 with $\sigma_0 = \mathrm{id}$). Hence:

- Γ = scalars: every θ works (M-SIDH, SILBE), but the Weil pairing leaks the square of the mask, leaving $2^{\omega(N)}$ candidates, so N needs at least 2λ prime factors.
- Γ = diagonal in (P, Q): works iff M is diagonal, i.e. (P, Q) is an eigenbasis of θ. The centraliser of a non-scalar θ mod N is exactly the diagonal torus, so diagonal masks are the **largest** mask set compatible with a single secret endomorphism.
- Once ψ_θ is known on $E_C[N]$ with $N^2 > \deg \psi_\theta = d^2 \cdot n(\theta)$, Kani's lemma gives ψ_θ as an explicit isogeny and $\ker \hat{\varphi} = \ker \psi_\theta \cap E_C[d]$ in the generic case.

Parity (found while implementing). Eigenlines of full order 2^e exist iff 2 splits in $\mathbb{Z}[\theta]$, i.e. $\mathrm{disc}\,\mathbb{Z}[\theta] \equiv 1 \bmod 8$, i.e. trace t odd and norm w even. Then one eigenvalue is even, $\psi_\theta + [c]$ has even degree for every c, and the Kani kernel degenerates on the 2-torsion. The fix is to decode with the trace-zero element $\theta' := 2\theta - t$, of norm $\Delta' := 4w - t^2 \equiv 7 \bmod 8$, which has the same eigenlines and eigenvalues $\pm s_0$ with $s_0^2 \equiv -\Delta' \bmod 2^{e+2}$. Then $\psi' := \varphi\theta'\hat{\varphi}$ has odd eigenvalues $\pm d \cdot s_0$.

3. The scheme

Parameters. $p = 2^e \cdot 3^f \cdot c - 1$, $N = 2^e$, $d = 3^f \approx 2^{2\lambda}$, e fixed by §5. All arithmetic is over $\mathbb{F}_{p^2}$.

KeyGen

- Walk from E_0 (j = 1728) by a secret isogeny of degree $> 2^{4\lambda}$ to a statistically uniform E_A; keep the connecting ideal, so $O_A = \mathrm{End}(E_A)$ is known (Deuring tooling: “Deuring for the people”, SQIsign2D ideal-to-isogeny).
- Enumerate short non-scalar $\theta \in O_A$ (LLL on the trace-zero lattice; shortest norm $\approx p^{2/3}$ for a random maximal order). Pick θ with (i) $\Delta' \equiv 7 \bmod 8$ and (ii) $N^2 - d^2\Delta' = c^2 + c'^2$ a sum of two squares (c even, c′ odd; e.g. a prime $\equiv 1 \bmod 4$, Cornacchia).
- Compute the eigenbasis (P, Q) of θ on $E_A[2^e]$ by evaluating θ on a basis and solving the 2×2 eigenproblem mod 2^e. Also fix a canonical basis (P_d, Q_d) of $E_A[3^f]$.
- pk = (E_A, P, Q). sk = (s_0, c, c'). Note that θ itself is not needed at decryption time.

Encaps

- Sample $s \in \mathbb{Z}/3^f$ and odd nonces $a, b \in \mathbb{Z}/2^e$ from a seed. $K := P_d + [s]Q_d$; $\varphi: E_A \to E_C := E_A/\langle K \rangle$ (3-isogeny chain).
- $ct = (E_C, S := [a]\varphi(P), T := [b]\varphi(Q))$; key = H(seed, ct).

Decaps

- ψ' acts on the ciphertext points by known scalars: $\psi'(S) = [d \cdot s_0]S$, $\psi'(T) = [-d \cdot s_0]T$, and $\hat{\psi}' = -\psi'$. (Derivation: $\psi'(a\varphi P) = a\varphi\theta'(\hat{\varphi}\varphi P) = a \cdot d \cdot \varphi(\theta' P) = a \cdot d \cdot s_0 \cdot \varphi(P)$; the nonces never enter.)
- Kani in dimension 2: with $\alpha := \psi' + [c]$ and $c^2 + c'^2 = N^2 - d^2\Delta'$, the matrix $F = [[\alpha, c'], [-c', \hat{\alpha}]]$ is an (N^2, N^2)-isogeny of $E_C{}^2$ with kernel $\{(c'x, -\alpha x) : x \in E_C[N^2]\}$, whose N-torsion part $\langle (c'S, -(ds_0+c)S), (c'T, -(c-ds_0)T) \rangle$ is known. Compute F as two N-halves from the actions of α and $\hat{\alpha}$; then $\alpha(X)$ is the first component of $F(X, 0)$ and $c'X$ the second, which also fixes the component isomorphisms of the split codomain.
- $\ker \hat{\varphi} = \ker(\alpha - c) \cap E_C[3^f]$ (cyclic of order d generically; otherwise try the few cyclic order-d subgroups). Recover φ, K and s; re-derive a, b from the seed, re-encrypt and compare (Fujisaki–Okamoto with implicit rejection).

Correctness. $\psi' = \varphi\theta'\hat{\varphi}$ is an endomorphism of E_C of degree $d^2\Delta'$ and trace 0, so $\deg \alpha = d^2\Delta' + c^2$ and $\deg \alpha + c'^2 = N^2$. Maximal isotropy of the kernel follows from $e(c'S, c'T) \cdot e(\alpha S, \alpha T) = e(S, T)^{c'^2 + \deg \alpha} = 1$. Because θ' is diagonal with odd eigenvalues it acts as the identity on $E_C[2]$; the kernel's 2-torsion is the diagonal of $E_C[2]^2$, so the (2,2)-chain necessarily starts with product-to-product steps before gluing into a Jacobian. Generic $\ker \varphi \not\subset \theta'$-stable subgroups gives $\ker \psi' \cap E_C[d] = \ker \hat{\varphi}$.

4. Prototype

A SageMath implementation of the full path lives in `research/eigen_seta_prototype_20261011/`. Because the FESTA and Castryck–Decru chain code assumes one gluing, Jacobian steps, and one split, a generic $(2^e, 2^e)$-chain evaluator (`kani_chain.py`) was written that handles product→product, gluing, Richelot and splitting steps in any order, reusing the Richelot/Mumford formulas of Pope–Oudompheng–Panny (MIT).

Two special-key variants are implemented. cm: E_A is the curve with CM by $\mathbb{Z}[(1+\sqrt{-7})/2]$ ($j = -3375$, supersingular for $p \equiv 3, 5, 6 \bmod 7$); $\theta = \omega$ of degree 2 ($t = \pm 1$, $w = 2$, $\Delta' = 7$). Its End is public, so it is insecure as a key but exercises the whole mechanism. walk: $E_A = \varepsilon(E_{cm})$ for a secret 5^k-isogeny ε and $\theta = \varepsilon\omega\hat{\varepsilon}$ ($\Delta' = 7 \cdot 25^k$): Castryck–Vercauteren's §5.3 backdoor used by its owner. Both run the single-chain regime ($N > \deg \psi'$). The random-curve variant of §3 is **not** implemented: it needs the two-halves evaluation (§5).

Results (toy sizes, correctness only)

variant	p (bits)	chain length e	sender degree	trials	correct	decaps time (Sage)
cm	41	26	3^5	25	25	≈ 0.2 s
walk, 5^2	45	29	3^4	25	25	≈ 0.2 s
walk, 5^2	55	36	3^5	25	25	≈ 0.3 s
walk, 5^3	66	44	3^6	10	10	≈ 0.8 s

In every trial the independent check of the lemma (direct evaluation of $\varphi\theta'\hat{\varphi}$ on the ciphertext points) matched the predicted eigenvalues $\pm d \cdot s_0$, and the chain began with two or three product-type steps before gluing, as predicted. Two corner cases were met and handled: a non-cyclic $\ker \psi' \cap E_C[3^f]$ (resolved by testing the cyclic order-d candidates against the codomain E_A) and a degenerate Mumford input in the splitting formulas for some payload points (resolved by retrying with a fresh basis of $E_C[3^f]$; about 1 in 25 trials; cause not isolated).

Experiment: do wrong eigenvalue guesses fail early?

An attacker who tries to recover θ from the public eigenbasis must guess Δ' (hence s_0, c, c′). Every pairing-consistent guess gives a maximal isotropic kernel, so the chain runs; the question is whether a wrong guess fails only at the final split or earlier (which would allow a digit-by-digit attack, as in Castryck–Decru). At e = 36 (walk) and e = 26 (cm), with 43 guesses each including guesses agreeing with the true Δ' on 3–13 low bits:

outcome	walk, e = 36	cm, e = 26
wrong guess ran the full chain and ended on a Jacobian (no split)	43 / 43	42 / 43
wrong guess split to $E_C \times E_C$ (false positive)	0	1 (random guess)
true value split to $E_C \times E_C$	yes	yes

No early-abort signal appeared. The single false positive is explained by the CM toy curve: E_C is 3^f-isogenous to a curve of discriminant -7, so other elements of the same quadratic order share the eigenbasis and some wrong Δ' are genuine. This is an artefact of the special key and, at the same time, a warning for special-key variants in general.

5. Parameters and the two-halves requirement

Constraints: meet-in-the-middle on the 3^f-isogeny forces $d \geq 2^{2\lambda}$; decoding needs $N > d\sqrt{\Delta'}$; for a random curve the shortest odd-trace endomorphism has norm $w \approx p^{2/3}$; and $p \approx N \cdot d$. Solving gives $N \approx d^2$, $p \approx d^3$. Derived figures at $\lambda = 128$:

quantity	value	comparison
sender degree $d = 3^f$	$3^{162} \approx 2^{257}$	FESTA uses 257–260-bit degrees
eigenbasis torsion $N = 2^e$	$e \approx 515$	$N > d \cdot p^{1/3}$

quantity	value	comparison
prime p	$\approx 2^{772}$	FESTA 1292, POKÉ 431, SILBE 13 013 bits
public key	≈ 325 bytes (curve + compressed eigenbasis)	FESTA 561, POKÉ 324
ciphertext	≈ 325 bytes (curve + compressed masked points)	FESTA 1122, POKÉ 648
decryption	one $(2^{515}, 2^{515})$-chain in dimension 2 as two halves + 3^{162} kernel recovery	FESTA-class cost

Why the two halves are essential. The table assumes the (N^2, N^2)-isogeny is computed from N-torsion as two N-halves ($N^2 > \deg$). With a single chain ($N > \deg$, the regime the prototype implements) the condition becomes $N > d^2\Delta' \approx 4d^2p^{2/3}$ with $N \le p$, which forces $p > d^9 \approx 2^{18\lambda}$; and for curves close to E_0 every odd-trace endomorphism has norm at least $p/(4D_A)$, which the prototype confirmed numerically, so a single chain never fits a random key. In Richelot/Mumford form the second half needs the dual chain plus an explicit isomorphism of the middle Jacobians; in theta coordinates (SQIsignHD, SQIsign2D) duals are explicit. This is the next implementation milestone.

Hidden-CM-curve variant at full size. Finding the secret walk ε costs $\sqrt{D_A}$ and finding $\varepsilon\omega\hat{\varepsilon}$ costs about $\sqrt{7}\cdot\sqrt{D_A}$, so $D_A \approx 2^{2\lambda}$, $\Delta' \approx 2^{4\lambda}$, $N \approx 2^{4\lambda}$, $p \approx 2^{6\lambda}$: the same prime size as the random-curve variant, a far cheaper key generation, but a different and probably weaker key-recovery assumption (a hidden small-discriminant orientation plus a walk, rather than EndRing).

6. Security analysis

Two assumptions are new and have no reduction to a studied problem.

- **Eigen-CIST** (message recovery): given $(E_A, P, Q, E_C, [a]\varphi P, [b]\varphi Q)$ with random odd a, b, find φ. The Weil pairing leaks $ab\cdot d \bmod 2^e$, leaving $\approx 2^{e-1}$ masks; an attacker needs the correct mask on a block $2^{e'}$ with $2^{2e'} > d$ before Robert's attack applies, cost $\approx \sqrt{d} \approx 2^{\lambda}$. This is FESTA's CIST problem family, unbroken since 2023 on generic bases, except that here the basis is a secret eigenbasis.
- **EndRing with an eigenbasis hint** (key recovery): given (E_A, P, Q), find θ. Guessing the eigenvalues modulo a block $2^{e'}$ with $2^{2e'} > \Delta'$ and running Kani costs $\approx \Delta' \approx 2^{2\lambda}$; cycle search for an endomorphism of norm Δ' costs $\approx 2^{2\lambda}$; generic EndRing costs $p^{1/2} \approx 2^{386}$ ($p^{1/3} \approx 2^{257}$ heuristically). Once θ is known, $\mathrm{End}(E_A)$ follows at $|\mathrm{disc}|^{1/4}$ classical or subexponential quantum cost (Herlédan Le Merdy–Wesolowski). The eigenlines $\langle P\rangle$, $\langle Q\rangle$ also expose two horizontal 2-walks of length e in the $\mathbb{Z}[\theta]$-orientation, a negligible fraction of the class group; self-pairing attacks (Castryck et al. 2023) need $m^2 \mid \Delta'$ with $m \mid N$ and are excluded by Δ' odd.

attack	best known cost at the §5 parameters	status
mask guessing + Robert (ciphertext)	$\approx 2^{128}$ classical; $\approx 2^{64}$ with Grover	same as FESTA and every masked-torsion scheme
meet-in-the-middle on the 3^{162}-isogeny	$\approx 2^{128}$	generic
eigenvalue guessing + Kani (key)	$\approx 2^{256}$	no early-abort signal at toy size (§4)
cycle search for θ	$\approx 2^{256}$	generic
generic EndRing	2^{386} / 2^{257} / quantum 2^{193}	not the binding constraint
hidden-CM variant: find the walk ε	$\approx \sqrt{D_A}$	sets $D_A \approx 2^{2\lambda}$; weaker assumption than EndRing

Level achieved. The prototype provides no security: its primes are 32–66 bits, the cm key has public End, and the walk keys are found by exhaustive search in seconds. The design at the §5 parameters has a conjectured classical level of 128 bits and, like every masked-torsion scheme,

about 64 bits against a quantum mask-guessing attack unless d is raised to $2^{4\lambda}$. The figure is a conjecture because the two assumptions above are unproven; the only evidence is the toy experiment of §4.

Other requirements. Decryption uses the secret s_0, so a chosen-ciphertext attacker with a decryption oracle learns about it unless the Fujisaki–Okamoto transform with re-encryption and implicit rejection is in place (compare the Moriya–Onuki adaptive attacks on FESTA). Masks must be fresh per ciphertext and derived from the seed; partial mask leakage (Das–Invernizzi–Kutas–Meers 2025) and fault attacks on torsion masking (Gilchrist–Lai–Meyer 2026) apply as to FESTA. If θ ever leaks through a side channel, security drops to the subexponential one-endomorphism bound: constant-time Kani code is mandatory from the start.

7. What is new, and what is not

- vs Séta: masks make the hint useless to Robert's generic attack; torsion requirement $N > d\sqrt{\Delta'}$ instead of $N > D^2$; no trapdoor-curve construction.
- vs SILBE: diagonal instead of scalar masks removes the $\omega(N) \geq 2\lambda$ constraint (13 013 → ≈ 772 bits); no F_p-rational E_0, no Frobenius; a shortest endomorphism replaces the degree-√p connecting isogeny. Lost: SILBE's public key is any curve with no extra data; ours leaks an eigenbasis.
- vs FESTA/QFESTA/POKÉ: the public key carries no torsion images of a receiver isogeny from a known-End curve, so key recovery is an EndRing-type problem rather than CIST or C-POKE; E_0 is not used at encryption time.
- vs Castryck–Vercauteren §5.3: the “backdoored basis” is the honest receiver's secret, so the basis-hashing countermeasure does not apply.

The scheme does **not** solve open Problem 7 of ePrint 2026/1431: its ciphertext assumption is still a torsion-hint assumption. It would be the first practical-size encryption scheme whose key-recovery problem is an EndRing variant with a leakage term rather than a torsion-hint problem, if the assumptions hold.

8. Open problems and next steps

- Implement the two-halves Kani evaluation (dual Richelot chain plus middle-Jacobian isomorphism, or a theta-model implementation) and the random-curve key generation (LLL on $\mathrm{End}(E_A)$ with explicit evaluation of a short endomorphism).
- Repeat the wrong-guess experiment on random-curve keys, where the CM false positive cannot occur, and at larger sizes.
- Study eigen-CIST against the level-structure reductions of De Feo–Fouotsa–Panny and the generalised class-group actions of Arpin et al., with the eigenbasis as the level structure.
- Parameter script and a C implementation; constant-time chain code; Fujisaki–Okamoto wrapper.
- Formalise Eigen-EndRing and eigen-CIST; prove OW-CPA ⇐ eigen-CIST and key recovery ⇐ Eigen-EndRing.

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